```
PUT tech_books4/_doc/2
{
  "title":"Elasticsearch for Java Developers"
}

PUT tech_books4/_doc/3
{
  "title":"Elastic Stack in Action"
}
```

Because the `title` field's data type is `search_as_you_type`, Elasticsearch creates a set of subfields called *n-grams* that partially match the user's search. The `_index_prefix` generates edge-n grams such as `[e, el, ela, elas, elast, elasti, elastic]` for the word *Elastic*. A 2-gram token is a shingle token filter producing two tokens, `["elasticsearch", "in"]` and `["in", "action"]"`, for the title *Elasticsearch in Action*. Similarly, a 3-gram is a shingled token filter generating tokens like `["elasticsearch for java", "for java developers"]` for the title *Elasticsearch for Java Developers*. (The source code is available with the book's files, if you want to try these examples.)

In addition to these n-grams, the root field (`title`) is indexed as is with the given or default analyzer. All other n-grams are produced by employing various shingle-token filters, as shown in table 4.5.

Table 4.5 Subfields created automatically by the engine

Fields	Explanation	Examples
`title`	The `title` field is indexed with either a chosen analyzer or the default analyzer if one is not chosen.	If a standard analyzer is used, the title is tokenized and normalized based on the standard analyzer's rules.
`title._2gram`	The `title` field's analyzer is customized with a shingle-token filter. The shingle size is set to 2 on this filter.	Generates two tokens for the given text. For example, the 2-grams for the title `"`*Elasticsearch in Action*`"` are `["elasticsearch", "in"],["in","action"].`
`title._3gram`	The `title` field's analyzer is customized with a shingle-token filter. The shingle size is set to 3 on this filter.	Generates three tokens for the given text. For example, the 3-grams for the title `"`*Elasticsearch for Java developers*`"` are `["elasticsearch","for","java"], ["for","java","developers"]`
`title._index prefix`	The analyzer of `title._3gram` is applied along with an edge n-gram token filter	Generates edge n-grams for the field `title._3grams`. For example, the `_index_prefix` generates the following edge n-grams for the world `"Elastic": [e, el, ela, elas, elast, elasti, elastic]`

Because these fields are created for us, searching on the field should return type-ahead suggestions, because the n-grams help produce them effectively. Let's create the search query as shown in the next listing.